

Matrix functions and stability

A dimension-independent polynomial norm bound from super-identical pseudospectra

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Topic: Polynomial matrix norms and pseudospectral information

Rating rationale: Existing information controls polynomial norms with a dimension-dependent constant; a uniform comparison would strengthen pseudospectral inference about nonnormal dynamics and polynomial matrix functions.

Context and notation

All matrices are complex and all norms are spectral norms. Write $s_1(M) \geq \dots \geq s_N(M)$ for the singular values of M . Define

$$A \sim_{\text{sip}} B \iff s_j(A - zI) = s_j(B - zI) \quad (z \in \mathbb{C}, 1 \leq j \leq N).$$

Define the sharp comparison constant

$$C_N = \sup \left\{ \frac{\|p(A)\|_2}{\|p(B)\|_2} : A, B \in \mathbb{C}^{N \times N}, A \sim_{\text{sip}} B, p \in \mathbb{C}[z], p(B) \neq 0 \right\}.$$

As recorded in [SP-15](#), super-identical pseudospectral data also imply ordinary similarity, so $p(A) = 0$ if and only if $p(B) = 0$.

Problem statement

Determine whether $\sup_{N \geq 1} C_N < \infty$. Equivalently, does there exist an absolute constant C such that

$$\|p(A)\|_2 \leq C \|p(B)\|_2$$

for every dimension, every super-identical-pseudospectral pair and every polynomial? Determining the sharp C_N is the closely related quantitative question, retained here rather than split into another entry.

References

- Fortier Bourque and Ransford, *Super-identical pseudospectra*, p. 513, immediately after Theorem 1.3, explicitly ask both whether their $\sqrt{N}$ bound is optimal and whether a dimension-independent bound exists. The sharp-constant notation above is editorial; the boundedness question is source-stated.
- Thomas Ransford and Nathan Walsh, *A four-mean theorem and its application to pseudospectra*, arXiv:2109.14472v2, 9 July 2022, Theorem 1.3, prove

$$\|p(A)\|_2 < \sqrt{N-2} \|p(B)\|_2 \quad (N \geq 4)$$

unless both sides vanish. **Proposition 5.1** and its following argument show sharpness for $N = 4$: $C_4 = \sqrt{2}$. **Theorem 1.4** concerns unbounded condition numbers of similarity transforms in fixed dimension, a different quantity.

Thus $C_N \leq \sqrt{N-2}$ for $N \geq 4$. The small-dimensional unitary/transpose classification gives $C_1 = C_2 = C_3 = 1$.

Scope and status check

The unknown absolute constant, not the superseded $\sqrt{N}$ sharpness guess, is the admission target. This comparison does not follow just from ordinary similarity, since a badly conditioned similarity can change norms substantially. Searches through 2026-09-11 for super-identical polynomial norm bounds, dimension-independent bounds, sharp constants and successors to the four-mean paper found no full solution. The current arXiv version history still ends at v2, 9 July 2022. This is an older-source question supported by bounded searches; no 2026 explicit open-status reaffirmation was located.